

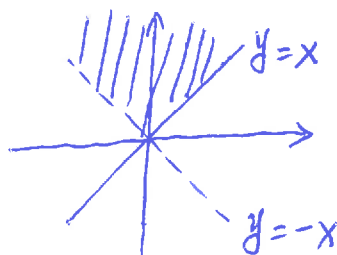
## Quiz 5

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

**Problem 1.** (4 points) Find and sketch the domain of the function:

$$f(x, y) = \sqrt{y-x} \ln(y+x).$$

$$\begin{cases} y-x \geq 0 \\ y+x > 0 \end{cases} \Rightarrow D = \{(x, y) \mid y \geq x \text{ and } y > -x\}$$



**Problem 2.** (6 points) Find the limit if it exists or show that the limit does not exist.

$$(a) \lim_{(x,y) \rightarrow (1,-1)} y^3 \cos(x+y)$$

$$= (-1)^3 \cos(1-1)$$

$$= -1 \cdot \cos 0$$

$$= -1$$

$$(b) \lim_{(x,y) \rightarrow (0,0)} \frac{y^3}{x^3 + 3y^3}$$

We examine the values along the line  $y=kx$  ( $x \neq 0$ )

$$\begin{aligned} \frac{y^3}{x^3 + 3y^3} &= \frac{(kx)^3}{x^3 + 3(kx)^3} = \frac{k^3 x^3}{(1+3k^3)x^3} \\ &= \frac{k^3}{1+3k^3} \end{aligned}$$

$$\text{When } k=1, \quad \frac{y^3}{x^3 + 3y^3} = \frac{1^3}{1+3 \cdot 1^3} = \frac{1}{4}$$

$$k=-1, \quad \frac{y^3}{x^3 + 3y^3} = \frac{(-1)^3}{1+3(-1)^3} = \frac{1}{2}$$

$\Rightarrow$  the limit does not exist.